Polar Coordinate Representations for DNA Sequence Compression: Why Geometry Alone Is Not Enough

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Abstract

We investigate whether mapping DNA bases to phases on the unit circle — a technique inspired by PolarQuant, a recent method for compressing transformer key–value caches — can reduce the storage footprint of genomic sequences below the 2 bits/base information-theoretic floor. We implement four increasingly sophisticated variants: a naive per-base polar stream, a block-summed polar representation, a keyframe–delta encoding in the style of video codecs, and a bit-packed version of the same. All variants are benchmarked against a 2-bit packed baseline and against general-purpose entropy coders (zlib, bz2, lzma) on three synthetic datasets: uniformly random DNA (100 kb), AT-rich DNA (100 kb), and highly repetitive DNA (100 kb). Our results show that the polar coordinate transform alone provides no compression benefit for random or weakly-biased sequences; in fact, unless carefully bit-packed, it inflates the data stream and makes downstream entropy coders perform worse than on the raw 2-bit representation. Only lossy variants that discard sequence order within a block achieve sub-baseline rates, and these cannot be used for storage of reference genomes. We conclude that polar mappings are a coordinate change, not a compression primitive, and that meaningful lossless gains on real DNA require statistical context models (e.g., order-k Markov predictors) rather than geometric transformations.

1. Introduction

Modern genomics generates sequence data at a rate that outpaces storage capacity. A single human genome in raw FASTA format occupies roughly 3 GB of plain text, or 750 MB if encoded at the 2-bits/base information-theoretic lower bound. Specialised tools such as *GeCo* and *CMIX* can push this below 1.6 bits/base by exploiting long-range statistical structure, but these pipelines are complex. Simpler, mathematically motivated transformations are therefore attractive — if they can deliver real savings.

PolarQuant (Hu et al., 2025) recently demonstrated that quantising the polar coordinates of key–value embedding vectors in transformer inference produces large storage savings with minimal quality loss. The intuition is that angles in high-dimensional caches concentrate near a small number of dominant directions, so even coarse angular quantisation preserves most of the information. This raises the natural question: does an analogous geometric trick work for DNA, where each of the four bases can be assigned a distinct phase on the unit circle?

This paper answers that question empirically. We map each base to a phase (A = 0, C = $\pi/2$, G = π, T = $3\pi/2$), build four increasingly sophisticated polar-based codecs, and compare each against two baselines: 2-bit packed encoding and general-purpose entropy coders applied to the raw ASCII sequence.

2. Background

2.1 Voss Mapping and Phase Representation

The Voss mapping assigns each nucleotide to a position on the unit circle. Using A = 0, C = $\pi/2$, G = π, T = $3\pi/2$, a sequence s■s■…s_n becomes a sequence of unit complex numbers $z_i = e^{i\phi(s_i)}$. This representation is commonly used in digital signal processing of genomes — for example, to detect tandem repeats via the discrete Fourier transform.

2.2 Shannon Bound for Uniform 4-ary Sources

For a sequence drawn uniformly from a 4-letter alphabet, the Shannon entropy is exactly 2 bits/symbol. No *lossless* encoder, regardless of cleverness, can compress random DNA below this floor. Any reported ratio above 1.0× against 2-bit packing on random data therefore implies either (a) non-uniform statistics in the test sequence, or (b) loss of information.

2.3 What Entropy Coders Do

zlib (Deflate), bz2 (Burrows–Wheeler + Huffman), and lzma (LZ77 + range coder) exploit redundancy by modelling symbol frequencies and repeated substrings. Their effectiveness is bounded by the entropy of the symbol stream they see. If we pre-process DNA in a way that *expands* the stream — for example, by turning a 2-bit symbol into an 8-bit byte — the entropy coder must first undo the expansion before it can reach the underlying 2-bit limit; it can never do better.

3. Methods

3.1 Codec Variants

We implemented four codecs of increasing complexity. All share the same Voss phase assignment.

- **V1 — Per-base polar stream:** emit one byte per base holding the quantised phase (8-bit precision). Lossless, 4× expansion before entropy coding.
- **V2 — Block-summed polar:** sum the complex values within a block of B bases and emit (magnitude, phase) at 5 and 8 bits respectively. Lossy; sequence order within a block is discarded.
- **V3 — Keyframe + delta:** for each block, emit the block centroid phase (an 8-bit keyframe) plus a 4-bit delta per base relative to the centroid. Lossless for $B = 16$, $delta\_bits = 4$; lossy for smaller delta budgets.
- **V3-packed:** same as V3 but two 4-bit deltas are packed into a single byte, eliminating the “byte tax” introduced by storing a 4-bit value in an 8-bit container.

3.2 Datasets

We evaluate on three synthetic sequences of 100 000 bases each: **Random** (uniform over {A,C,G,T}), **AT-rich** (P(A) = P(T) = 0.4; P(C) = P(G) = 0.1, modelling bacterial extremes), and **Repetitive** (the motif GATTACA tiled to fill 100 000 bases).

3.3 Evaluation Protocol

For each (codec, dataset) pair we compute the compressed size under zlib-9, bz2-9, and lzma preset-9, and report the resulting bits/base and compression ratio relative to the 2-bit packed baseline. For lossless codecs we additionally verify round-trip integrity via a decoder. Code and data are available in the accompanying repository.

4. Results

4.1 Entropy Invariance Under Polar Mapping

We first confirmed a sanity check: Shannon entropy of the raw ASCII sequence, the 8-bit polar quantisation, and the 4-bit polar quantisation are numerically identical (1.9880 bits/symbol on a human mitochondrial DNA sample). This is expected — a bijective mapping between the alphabet {A,C,G,T} and four distinct phase bins cannot change the empirical distribution.

4.2 Block-level Compression on Three Datasets

Table 1 summarises compressed sizes on 100 000-base sequences. All sizes are in bytes; bits/base is computed as 8 · bytes / n. The baseline is 25 000 bytes (2-bit packing).

Method	Random	AT-rich	Repetitive
2-bit packed (baseline)	25 000 (2.00)	25 000 (2.00)	24 998 (2.00)
Raw ASCII + bz2	27 524 (2.20)	26 044 (2.08)	58 (0.005)
Raw ASCII + lzma	28 092 (2.25)	25 004 (2.00)	156 (0.012)
Polar V1 (1 byte/base) + bz2	13 608 (2.18)*	13 055 (2.09)*	58 (0.009)*
Polar V3-packed + bz2	39 559 (3.17)	36 531 (2.92)	171 (0.014)
Polar V3-packed + lzma	42 352 (3.39)	37 908 (3.03)	196 (0.016)

Table 1. Bytes and (bits/base) after entropy coding. *V1 values from the 50 kb benchmark in our earlier runs; trends are identical at 100 kb.

4.3 Key Observations

Three observations stand out. **First**, on random DNA no polar variant beats the 2-bit baseline. The best lossless polar result is 2.18 bits/base (V1 + bz2), worse than the 2.00-bit floor.

Second, the V3 keyframe-plus-delta codec, despite its video-codec-inspired structure, actually *hurts* compression on random and AT-rich data (3.0–3.4 bits/base). The reason: a 4-bit delta carries only 2 bits of real information (which of four phase bins relative to the centroid), yet occupies 8 bits on disk when packed naively and is still bounded by the Shannon limit after packing. The additional keyframe overhead and the residual redundancy between the keyframe byte stream and the delta byte stream confuses zlib and lzma's models.

Third, the spectacular ratios on the repetitive dataset (> 100×) are *not* attributable to the polar mapping. Raw ASCII + bz2 compresses the same sequence to 58 bytes (431×). The polar pre-processor is essentially neutral here — the entropy coder does all the work.

4.4 Losslessness

For lossless variants (V1; V3 with B = 16 and delta_bits = 4) our decoder reconstructed the original sequence with zero mismatches on all three datasets. Reducing delta_bits below 4 at B = 16 introduced mismatches: delta_bits = 2 produced 75% errors on GATTACA repeat because the four base-phase quadrants cannot be resolved with only 4 angular bins shared around the centroid.

5. Discussion

5.1 Why Coordinate Changes Do Not Compress

Polar coordinates are a bijective reparametrisation of complex numbers. Applied to a stream of discrete symbols drawn from a four-letter alphabet, they preserve Shannon entropy exactly — our measurements in §4.1 confirm this. A compressor can only save bits by (a) modelling the *distribution* of symbols and spending shorter codes on more frequent symbols, or (b) modelling *context* to predict upcoming symbols. Polar mappings do neither.

5.2 Where Polar Representations Are Useful

This does not mean polar coordinates are useless in genomics. They are a standard building block for:

- **Signal-processing analyses** of periodic motifs (tandem repeats, codon periodicity via the DFT on the phase stream).
- **Similarity sketching**, where block-summed complex vectors can serve as locality-sensitive hashes for fast approximate search.
- **Lossy compression of aggregate statistics** — e.g., storing block-level GC-content and strand bias at reduced precision for downstream analytics, not for sequence storage.

5.3 What Would Actually Beat the Baseline

Real lossless DNA compressors beat 2 bits/base by exploiting *higher-order statistics*: an order-k Markov model predicts base $k+1$ from the previous k and feeds the predicted distribution to an arithmetic coder. On the human genome this drops entropy to roughly 1.6–1.7 bits/base. Adding reference-based encoding (storing only differences from a reference genome) pushes this below 0.5 bits/base. A polar pre-processor could be layered *on top of* such a context model, but would contribute only overhead unless the polar representation itself exposes structure that the context model cannot.

6. Conclusion

Mapping DNA bases to phases on the unit circle is elegant and conceptually appealing, but our empirical study shows it provides no lossless compression benefit over standard 2-bit packing for random or weakly-biased sequences. The keyframe-plus-delta variant inspired by video codecs actively hurts compression by expanding a 2-bit symbol into an 8-bit byte that entropy coders must subsequently squeeze back down. Apparent wins on repetitive data are attributable to the entropy coder alone, not to the polar transform. We conclude that polar representations are best viewed as a *coordinate change* — a tool for signal-processing analyses and lossy sketching — rather than as a compression primitive. Lossless DNA compressors that improve on the 2-bit floor must operate on the statistical structure of the sequence, via context modelling and arithmetic coding, not on its coordinate system.

References

[1] Hu, Y., et al. (2025). *PolarQuant: Quantizing KV Caches with Polar Coordinates.* arXiv:2502.02617.

[2] Voss, R. F. (1992). Evolution of long-range fractal correlations and 1/f noise in DNA base sequences. *Physical Review Letters*, 68(25), 3805.

[3] Pinho, A. J., & Pratas, D. (2014). MFCompress: a compression tool for FASTA and multi-FASTA data. *Bioinformatics*, 30(1), 117–118.

[4] Hosseini, M., Pratas, D., & Pinho, A. J. (2016). A survey on data compression methods for biological sequences. *Information*, 7(4), 56.

[5] Mahoney, M. (2013). *Data Compression Explained.* Dell, Inc. (Sections on context mixing and arithmetic coding.)

[6] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.